PREPBENCH / PRODUCT ARCHITECTURE PROPOSAL

The Search-First & Action-Ranked UI Architecture

Author: Antigravity PM Team | **Date**: September 2, 2026 | **Target**: PrepBench Next-Gen UI

1. Executive Thesis: Dissolving the Taxonomy Trap

For four sessions, the product debate centered on whether PrepBench should organize its interface by **Formats** (Exams, Interviews, System Design) or by **Subjects** (Scrum, Databricks, AI). This proposal asserts that **the two-axis debate is a false dilemma** created entirely by trying to build a traditional navigation tree.

The Real Google Move: Google's core architectural signature is not hierarchical taxonomies—it is *refusing to make the user navigate*. Gmail killed folders with search + labels. Drive is a search box with a filesystem attached. Power users live in command palettes. By establishing a single global Search / Command Input (Cmd + K) as the front door, the subject-vs-format debate completely dissolves: searching 'cost' or 'scrum events' instantaneously returns exams, design reviews, guide passages, and personal accuracy in one surface.

2. Explicit Assumptions & Risks (For Counter-Argument)

To enable rigorous counter-debate, the core assumptions underlying this design proposal are explicitly stated below alongside their vulnerabilities:

Assumption	Stated Rationale	Counter-Argument / Vulnerability
1. Session Intent	Users open the app with a specific topic or keyword in mind (e.g. *'I want to study Delta Lake'*).	**Vulnerability:** Novice users or passive learners may suffer from 'empty text box paralysis' when they don't know what keyword to type.
2. Command Palette Preference	Target users are technical engineers who prefer keyboard shortcuts (Cmd+K) over clicking sidebar links.	**Vulnerability:** Mouse-driven users may ignore keyboard shortcuts entirely if visual affordances aren't prominent.
3. Action-Ranked Home	Users prefer 3 ranked actionable recommendations (Resume, SM-2 Review, Weak Spot) over passive KPI charts.	**Vulnerability:** Users who want a birds-eye summary of overall certification readiness may feel deprived of visual progress gauges.
4. SM-2 Spaced Repetition Wiring	Surfacing the existing sm2_service.py backend engine on Home will drive daily short reviews.	**Vulnerability:** If a user misses 2 weeks, a 150-item SM-2 backlog could create overwhelming study debt.
5. Inline AI vs Drawer AI	Placing the [Why?] button directly inline on incorrect questions reduces friction compared to a side-drawer.	**Vulnerability:** Inline AI expansions can clutter the quiz card layout if detailed multi-paragraph explanations are generated.

3. UI Architecture & Wireframes

A. Home Screen: Search-First & Action-Ranked (3 Action Cards)

```
■ ■  Search anything or press Cmd + K (e.g. 'scrum events', 'delta lake', 'cost')...  ■
■                                                                                   ■
■ 1. ■■ RESUME SESSION                                                               ■
■ Databricks Design Review #3 ■■ Step 4 of 6 (Last active 18h ago)                  ■
■ [ ■■ Continue Design Review ]                                                      ■
■                                                                                   ■
■ 2. ■ DUE FOR SPACED REPETITION REVIEW (Powered by sm2_service.py)                 ■
■ 14 questions queued based on your memory decay curve (Scrum Events & Delta Lake)  ■
■ [ ■ Start 5-Minute Spaced Review ]                                                ■
■                                                                                   ■
■ 3. ■ WEAKEST TOPIC DRILL                                                         ■
■ Storage vs Compute Cost Optimization (38% accuracy across 3 attempts)             ■
■ [ ■ Target This Weakness ]                                                       ■
■                                                                                   ■
```

B. Command Palette Search Overlay (`Cmd + K`)

```
■ ■ Query: 'cost'                                                                  ■
■  ■ DESIGN REVIEW : Databricks FinOps & DBU Cost Optimization (Case Study #4)     ■
■  ■ EXAM QUESTIONS: 5 Questions on DBU Pricing & Cluster Autoscaling               ■
■  ■■ ROADMAP TOPIC : Databricks Governance & Cost Management (Phase 3)             ■
■  ■ YOUR ACCURACY  : 38% Accuracy on Cost Questions (3 Missed)                     ■
```

C. Practice Surface with Inline AI at Failure Point

```
■ Question 3 of 10  •  Databricks FinOps                                          ■
■ What is the primary advantage of Auto-terminating single-node clusters?          ■
■  [ ] A. It increases Spark driver memory allocation.                             ■
■  [X] B. It prevents idle DBU consumption when developer notebooks are inactive. (YOUR)  ■
■  [✓] C. It eliminates control plane management overhead. (CORRECT)               ■
■  ■ INLINE AI EXPLANATION (One-Click Inline Expansion)                           ■
■  Auto-termination stops compute charges, but control plane management remains managed ■
■  by Databricks. Option C accurately reflects architectural boundaries under DBU rules. ■
```

4. Backend Engine Integration Plan

A critical finding during this audit: **sm2_service.py and the SpacedRepetition model exist in the backend codebase, but have zero API router endpoints or frontend connections.** The mathematical engine is already built—it simply requires wiring.

Engine Component	Current Backend State	Proposed Surfacing
SM-2 Spaced Repetition	Fully implemented in app/services/sm2_service.py with ease factor floors (1.3) and interval math.	Wire to /api/v1/spaced-repetition/due and surface directly as Card #2 on the Home screen.
Weak Spot Detection	Calculated inside analytics_repository.py and system_design_service.py.	Surface as Card #3 on Home: 1-click targeted drill for the user's lowest accuracy topic.
LLM Gateway	Abstracted in app/llm/gateway.py with fallback to local env / Gemini.	Wire directly to inline [Why?] feedback buttons on quiz failure cards.

5. Empirical Telemetry Plan (2-Week Usage Query)

Rather than debating taxonomy theoretically, PrepBench can settle the Subject vs. Format question empirically. Every practice attempt is logged in SQLite (exam_simulator.db) with timestamps, domains, and session modes. Running the following SQL query after 2 weeks of natural usage will provide definitive evidence:

```sql
SELECT
    q.domain,
    s.exam_mode,
    COUNT(*) as total_attempts
FROM exam_answers a
JOIN questions q ON a.question_id = q.id
JOIN exam_sessions s ON a.session_id = s.id
GROUP BY q.domain, s.exam_mode;
```

Decision Metric: If usage clusters heavily by domain across multiple modes in a single sitting, subject-first scoping is empirically validated. If usage clusters by exam mode across multiple domains, format-first navigation is validated.